Sample mean, σ unknown -> Student t

1 RECOGNISE IT

- Probability concerns $\bar{X}$.
- Population σ is unknown and sample s is supplied.
- Population is normal/approximately normal under the course conditions.

2 DECISION / METHOD

1. Use s/√n as estimated SE.
2. Set df=n-1.
3. Convert boundaries to t.
4. Use T.DIST and the correct tail/difference.

3 FORMULA

$t=(\bar{x}-\mu)/(s/\sqrt{n})$

$df=n-1$

4 FORMULA SHEET VS WHAT I KNOW

Gives: s/√n, $t_{(n-1)}$ when σ is unknown.
Still know: normality condition, df, and tail handling.

5 EXCEL ROUTE

- Left: `=T.DIST(tcalc,df,TRUE)`
- Right: `=1-T.DIST(tcalc,df,TRUE)`
- Between: upper T.DIST minus lower T.DIST.

6 TINY WORKED EXAMPLE

μ=30, s=5, n=25. For P($\bar{X}$<32), `t=(32-30)/(5/SQRT(25))`=2, df=24; use T.DIST(2,24,TRUE).

7 TOP EXAM TRAPS

- Using Z because n looks large while σ is unknown in this course lane.
- df=n.
- Using σ notation for sample s.

8 SAY ALOUD

Unknown σ means estimate the standard error with s and use Student t with n-1 degrees of freedom.

9 EXIT CHECK

Which Excel function gives a cumulative-left t probability?

ONE-GLANCE VISUAL ROUTE

